

Identity-Centered User-Generated Content to Create Occupational Identity with Visual Computing

Tim McLaughlin

School of Performance, Visualization
& Fine Arts, Texas A&M University,
USA

timm@tamu.edu

Tracey L. Moore

School of Architecture, Prairie View
A&M University, USA
tymoore@pvamu.edu

Aaron P. Thibault

The Gearbox Entertainment Co., USA
aaron.thibault@gearbox.com

Kim B. Wright

Kindler Institute for Urban Research,
Rice University, USA
kimwright@rice.edu

Hira T. Roberts

School of Architecture, Prairie View
A&M University, USA
hitariq@pvamu.edu

Jinsil H. Seo

School of Performance, Visualization
& Fine Arts, Texas A&M University,
USA

hwaryoung@tamu.edu

Jason M. Orsatti

The Gearbox Entertainment Co., USA
jason.orsatti@gearbox.com

ABSTRACT

This talk reports on an approach to computer science education for pre- and early-adolescents in which the goal is the formation of occupational identity with visual computing developed collaboratively between a large game studio, a research-intensive university, and a historically black college/university (HBCU). This ongoing project takes place in a rural public-school setting in the United States. Our project is structured around the idea that identity-centered user-content creation projects can positively influence student self-professed performance and interest in science, technology, engineering, art, and math (STEAM) related subjects and interest in STEAM careers. The projects in our curriculum engage students in real-time 3DCG coding and asset creation activities commonly associated with game development. We describe the process of working with school administrators and teachers to create a technology-infused environment in which remote external partners play a collaborative role in curriculum development and delivery. This work has significance for efforts to remotely engage with rural students at an age when occupational identity development is forming and thus develop a potential to expand the pathway for underrepresented minorities.

ACM Reference Format:

Tim McLaughlin, Tracey L. Moore, Aaron P. Thibault, Kim B. Wright, Hira T. Roberts, Jinsil H. Seo, and Jason M. Orsatti. 2023. Identity-Centered User-Generated Content to Create Occupational Identity with Visual Computing. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Educator's Forum (SIGGRAPH '23 Educator's Forum)*, August

06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages.
<https://doi.org/10.1145/3587424.3595583>

1 INTRODUCTION

Employment opportunities in computer science are growing faster than the average for all other occupations and the demand for knowledge and skills in visual computing – computer science plus overlapping skills in areas such as human-computer interaction, design, and technical art – has grown 601% faster than the computer science market overall [Burning Glass Technologies, 2019]. Real-time 3D graphics (3DCG), enabling immersion, data-driven operations, and interactivity, though popularized through entertainment media, is growing fast across the U.S. economy. 3DCG is being incorporated into the development, design, training, testing, and decision-making processes of industries including manufacturing, architecture, engineering design, military, and medical [U.S. Bureau of Labor Statistics, 2022]. This provides both an opportunity and a charge to broaden the workforce pathway and re-imagine how visual computing can be taught, more accessible, and appealing to students of color for whom the pathway to STEM professions is often less accessible.

The strategy used in this project is called identity-centered user-generated content (ICUC). ICUC is intended to spark an occupational interest in the fields of visual computing by providing tangible connections between digital assets, the student, and the development of the game. Near-Peer Mentors (NPMs) play a key role in providing a sense of accessibility and projects that incorporate student identity to promote a sense of belonging.

2 BACKGROUND AND RELATED WORK

Despite the fast growth rate in computer science and visual computing employment opportunities, and though participation in STEM fields by Blacks and Hispanics has risen over the past 10 years, it is still significantly out of alignment relative to representation in the overall U.S. population [National Science Foundation, 2019].

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Educator's Forum, August 06–10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0146-7/23/08.

<https://doi.org/10.1145/3587424.3595583>

Thoughtfully designed interventions provide a method of addressing these issues and can have long lasting effects. Significantly, the effects of the interventions are more often found to be stronger, more beneficial, and longer lasting in the URM members of the groups studied [Yeager et al., 2014]. Our approach focused on 5th – 8th graders. Adolescence is the period in the life course when young people start to ask the question of “who am I?” The answer to this question will continue to evolve over one’s lifetime, but it begins in adolescence and forms the basis of one’s sense of self [McNeely and Blanchard, 2010]. Having the ability to create content and artifacts furthers creative expression for future and emerging identities [Ellison, 2017]. Popular platforms such as *Minecraft* and *Roblox* have introduced many students in this age range to the ideas of user-generated content to create narratives, and game modding has demonstrated the capacity to develop interest in computer science because it encourages the development of critical thinking skills, problem solving, computational thinking, and creativity [Grizioti and Kynigos, 2018]. Several recent studies have demonstrated identity-oriented impacts linked to art modding, primarily around avatar design [Linden Research, 2011] [Numedeon, 2023].

3 ICUC IN SCHOOLS

This project was set in a large public independent school district (ISD) in Longview, Texas, a semi-rural area of the U.S. The student population is among the most diverse in the area with a demographic of Hispanic (40%), Black (35%), White students (20%), and Asian students (1.5%). Seventy-four percent of students come from economically disadvantaged households. Prior to initiating the classroom intervention, several steps were undertaken in preparation, relationship building, and consultation that were key in developing the outreach project’s exercises that the students would use to create ICUC. All parties involved –industry partners from Gearbox Entertainment, administrators and teachers from LISD, and NPMs and project leaders from Texas A&M University and Prairie View A&M University – held multiple consultations via video conferencing and site visits to understand the project relative to existing curricula and the classroom and technology environment for teachers and students. The curriculum is composed of three successive projects, Projects 0, 1, and 2, with Project 0 intended for the 5th grade students and Projects 1 and 2 intended for the 6th, 7th, and 8th grade students. It is within the design and implementation of each project that collaborative flexibility of process and delivery occurs. Flexibility is not only important to promote a collaborative spirit and sense of agency with the teachers and NPMs, but also to properly ground expectations within the logistics, technology, teacher preparedness and other variables on site at each campus. The goal was to successfully merge the aims of an existing curriculum with the realities in the classroom and marry those two with the project’s objectives of creating ICUC.

The ICUC curriculum for Project 0 was delivered at two elementary school campuses to 5th graders (ages 10–11). It took the form of students learning visual block coding through *Scratch* and *CodeCombat*. From the ICUC perspective, students were expected to create characters that resemble themselves or someone close to them and create environments that resemble a place of familiarity. The ICUC curriculum for Project 1 was approached at Middle

School level (ages 11–14) in the form of avatar, prop and environment design and creation in a digital content creation tool (DCC) for incorporation into a game engine, the *Unreal Engine* in our case. This was approached as two separate methods to offer a range of skill development, self-expression and to accommodate limitations in terms of technology and accessibility. The first method was customization of an *Among Us* (AU) character through direct modeling techniques which the students created from a template AU 3D model. Students were instructed and given agency to customize the hair, clothes, and physique of their template character to properly reflect their own temperament and personalities. Students are also required to write a design document explaining their design decisions. A second part of Project 1, yet to be implemented, is the development of a prop and environment through photogrammetry and photo modeling.

The lessons that we teach as they relate to game design and development, have their core in the practices of allowing students to recognize and develop their own personal identities. This is done by teaching students the process necessary to create assets of their own through techniques, such as photogrammetry, and instructing them to make assets of the things around them and those that matter to them, as well. In doing so, students can develop critical thinking and agency as they introduce something new into a game environment. Though this process, in itself, is straightforward, understanding the impact that ICUC can have on these students could give us insight into how this can be an effective method for helping to foster occupational identity in visual computing. Representation matters, especially for students in underrepresented groups. One of the goals of this project is to provide students with a pathway to seeing themselves in visual computing fields and careers -the assets they create directly imported to games they modded. This intimate relationship with the work could inspire and influence an occupational interest in visual computing.

ACKNOWLEDGMENTS

Funding provided by the Simons Foundation’s Science Sandbox. Thanks to our partners: the teachers and administrators at Longview Independent School District in Longview, Texas, USA.

REFERENCES

- Burning Glass Technologies. 2019. Visualizing the Future: Demand for 3D Graphics and Real-time 3D Across the Economy. https://www.burning-glass.com/wp-content/uploads/Visualizing_future_3D_skills.pdf
- U.S. Bureau of Labor Statistics. 2022. Occupational Outlook Handbook: Computer and Information Technology <https://www.bls.gov/ooh/computer-and-information-technology/home.htm>.
- National Science Foundation. 2019. The STEM Labor Force of Today: Scientists, Engineers, and Skilled Technical Workers. <https://ncses.nsf.gov/pubs/>
- D. S. Yeager, V. Purdie-Vaughns, J. Garcia, N. Apfel, P. Brzustoski, A. Master, W. T. Hesser, M. E. Williams, and G. L. Cohen. 2014. Breaking the cycle of mistrust: Wise interventions to provide critical feedback across the racial divide.” *J. Exp. Psychol. Gen.*, vol. 143, no. 2, pp. 804–824.
- C. McNeely and J. Blanchard. 2010. *The Teen Years Explained: A Guide to Healthy Adolescent Development*. Jayne Blanchard Publishing.
- T. L. Ellison. 2017. Digital Participation, Agency, and Choice: An African American Youth’s Digital Storytelling About *Minecraft*. *J. Adolesc. Adult Lit.*, vol. 61, no. 1, pp. 25–35.
- M. Grizioti and C. Kynigos. 2018. Game modding for computational thinking: an integrated design approach. Proceedings of the 17th ACM Conference on Interaction Design and Children, New York, NY, USA, pp. 687–692.
- Linden Research, Inc. 2011. “Teen Second Life,” *Second Life Wiki*. https://wiki.secondlife.com/wiki/Teen_Second_Life
- Numedeon, Inc. 2023. “Welcome to Whyville.” <http://whyville.net/smmk/nice>